UNIVERSITY of York

School for Business and Society

Module Title: Quantitative Analysis in Business Studies	**Module Code:** MAN00035C
Module Leader: Yuning Li, and Alexandra Dias	**Submission Deadline:** 11:00:00am UK Time, Wednesday 6th May 2026
Maximum Word Count: 1,200 words	**Weighting:** 50%
Estimated feedback return: This will normally be within 25 working days of the initial submission.	

Contents of this assessment brief

1. Academic Regulations

You should note that:

- **Submission deadline:** Work must be submitted by the dates specified. Where work is submitted after the deadline the following **penalties** will apply:
  - o Up to one hour after the submission deadline, a penalty of five marks
  - o Over one hour late but within 24 hours of the submission deadline, a penalty of ten marks
  - o After the first 24 hours of the submission deadline, ten marks will be deducted for every 24 hours (or part thereof) that the submission is late for a total of five days, after which it is treated as a non-submission and given a zero mark.

2. If things go wrong (exceptional circumstances and self certification)

- If you are unable to complete your open assessment by the submission deadline, you can **apply for an extension** either through the exceptional circumstances process or through self-certification. **Applications must be made before the deadline**. For further information and the claim forms, please refer to:
    - Exceptional Circumstances affecting assessments (for information on self-certification for assessments, please scroll to the bottom of this webpage)

- If you submit your open assessment on time but feel that your performance has been affected by exceptional circumstances, you may submit a retrospective Exceptional Circumstances Affecting Assessment claim form within 7 days after the published assessment submission deadline. If you do not submit by the deadline indicated without good reason, your claim will not be considered.

Please take proper precautions to safeguard your work and **remember to make backup copies of your data.** The University provides all its students with storage space on the University server, and you should save and back up any work in progress on this server on a regular basis. **Computer failure and theft of your equipment or storage media are not considered exceptional circumstances and extensions cannot be granted for work lost for these reasons**.

3. Generative AI Guidance for this assessment

Artificial Intelligence (AI) models can be a powerful tool to support your learning. The University has guidance on the use of AI in assessments. This details appropriate and inappropriate use of digital tools.

As per the academic regulations, in all cases you must submit work that is your own, acknowledging any part of it that has been informed by another source – including content that is AI generated. By submitting your work, you are agreeing to the following statement:

I confirm that this assignment is my own work, except where I have acknowledged the use of works from other sources, including the use of any artificial intelligence (AI) tools, in accordance with what is allowable as described in the assessment brief.

The following information provides specific guidance for this assessment about what level of AI use is appropriate for this assessment (the table below indicates the level of GAI integration):

Please note the following:

- AI should not be used as a substitute for your own knowledge, and you should never include any material that you do not understand and could not explain if asked.
- Not being able to explain your work when asked is likely to be a key factor when considering cases of academic misconduct related to AI.

Level of GAI - integration	Description	Y/N
No GAI	**GAI must not be used** *(AI tools embedded within software for structure suggestions, checking spelling, grammar and referencing **may be used,** AI **cannot be used** to translate work)* Your assignment should be produced using information sourced by you from your learning materials and academic sources and cited appropriately.	No*
GAI – Assisted idea development	**Students are not allowed to create content in submitted assessments using GAI** *(AI tools embedded within software for structure suggestions, checking spelling, grammar and referencing **may be used,** AI **cannot be used** to translate work)*; GAI can **only** be used for pre-assessment tasks such as for brainstorming, idea generation, and structuring.	Yes
GAI – assisted editing	**Students can use GAI to edit and enhance the clarity and quality of their work (***AI tools embedded within software for structure suggestions, checking spelling, grammar and referencing **may be used,** AI **cannot be used** to translate work)*;**.** Students will write the full first draft of an assessment and GAI is used to edit this draft	No*
GAI - Task Completion, Human Evaluation	**Students are permitted to use GAI for some or most elements of the task and demonstrate effective use of AI (***AI tools embedded within software for structure suggestions, checking spelling, grammar and referencing **may be used,** AI **cannot be used** to translate work)*	No*
Full - GAI	**Students are expected to use GAI creatively and collaboratively to complete the task and demonstrate effective use of AI including the use of AI to translate work**	No*

4. Module Learning Outcomes assessed

The following learning outcomes are assessed in this task:

Subject Content
- Describe various types of data, interpret descriptive statistics
- Explain and work with the concept of probability and distributions
- Describe a variety of sampling techniques, work with sample statistics, and confidence intervals
- State and test basic statistical hypotheses

Academic and graduate skills
- Accurately collect, handle and analyse data
- Apply various quantitative methods in a logical, rigorous, and concise way;
- Apply strict logical reasoning from assumptions to conclusion;
- Critically assess assumptions necessary to draw certain conclusions.

5. How will this assessment be marked?

The assessment will be marked using the assessment criteria relevant to your level of study.

How is my work graded

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes here.

6. <u>Assignment tasks:</u> Answer all the questions below

Question 1:

Consider the data sample provided in Table 1 concerning eleven firms listed in the London Stock Exchange and answer the questions that follow.

Table 1:

Firm code	Industry	Share Price (£)	Bond Credit Rating	Number of Outstanding Shares	Defaulted (Yes/No)
SWA	Banking	98.54	AA	12,971	No
BPT	Real Estate	79.56	CC	15,099	No
CGB	Banking	83.67	BBB	30,796	No
DKD	Real Estate	95.62	CCC	35,020	No
PLE	Healthcare	102.23	BBB	43,185	No
FIN	Healthcare	95.67	A	17,068	No
GGB	Insurance	189.43	AAA	43,185	No
PTH	Insurance	93.87	B	10,802	No
GRI	Banking	673.12	AA	32,924	No
SPR	Banking	89.30	BB	15,098	No
ITJ	Insurance	87.89	BBB	29,251	No

Answer the following questions:

a) For each column of Table 1 except the Firm code, state the type of the data contained in that column, explaining your answer in each case.

[5 marks]

b) Which measure of centrality would you choose in Table 1 for the
   i. Share Price data? Justify your answer.
   ii. Bond Credit Rating data? Justify your answer.

[4 marks]

c) Present an appropriate graph showing the proportion of firms in each industry sector. Justify the type of graph you chose to draw.

[5 marks]

d) Calculate, showing all workings, three different measures of spread (or dispersion) for the number of outstanding shares.

[6 marks]
Total: 20 marks

Question 2:

An international bank gives mortgage loan contracts to people who want to buy a house. In order to diversify the risk of borrowers not paying back the loan (or defaulting), the bank operates in three different countries: Japan, Brazil and Cape Verde. Last year, the bank loaned money to 800 clients in Japan, 500 clients in Brazil, and 100 clients in Cape Verde and the number of clients in Table 2 has independently defaulted.

Table 2:

Country	**Japan**	**Brazil**	**Cape Verde**
Number of clients	800	500	100
Number of defaulting clients	150	92	45

Answer the following questions:

a) Past experience indicates that clients in different countries default independently. What is the probability that this year:
   i. A randomly selected client in Cape Verde defaults? (show all workings)
   ii. Three randomly selected clients in Japan default and two randomly selected clients in Brazil default? (show all workings)

[5 marks]

b) A huge hurricane in the Atlantic Ocean hits both Brazil and Cape Verde leading to wide unemployment and the probability of default in each country to double. In addition, client defaults in both countries are now dependent such that the probability that a client in Brazil and a client in Cape Verde both default is 0.1. What is the probability that a client defaults in Brazil given that another client defaulted in Cape Verde? Justify your answer.

[10 marks]

Total: 15 marks

Question 3:

An investor decides to buy shares from firm GoodBet. The investor observes the stock price once a day. Based on historical data, the price of the GoodBet stock can either go up or go down from one day to the next. During the last trading year (250 days) the stock price went up in 112 days. The investor believes that past data is a good indicator of the future and that price movements in consecutive days are independent.

Answer the following questions:

a) What is the probability of the GoodBet stock going down on any given day? Explain your answer.

[5 marks]

b) What probability distribution would you use to model the number of days on which the GoodBet stock price goes up during the next year? Justify your answer.

[5 marks]

c) What is the probability of the GoodBet price going up on exactly 87 days during next trading year?

[5 marks]

d) Assuming that the investor measures the risk of their investment using the standard deviation of the model selected in part (b), calculate the risk using the available historical data.

[5 marks]
Total: 20 marks

Question 4:

Assume that you want to study the distribution of wages in the U.K. You don't have access to the wages for all the population in the U.K. Hence, you need to use sampling.

Answer the following questions:

a) For each of the following sampling strategies, state whether they would provide a representative sample of UK wages. Explain your answer in each case.

i. Collect the wage of the highest earner in each household in the U.K.
ii. Collect the wage of the 10 people who live closer to you.
iii. Collect the wage of one person randomly chosen in each of the 32 boroughs of London.
iv. Collect the wage by emailing 2,000 randomly chosen people in the U.K.
v. Divide the U.K. population into equal-sized groups based on gender, postcode and age. Collect wage data for 5 randomly chosen people in each group.

[10 marks]

b) Assume that from a representative sample of 10,000 independent individuals of the population you obtain a sample mean of £32,000 and a sample standard deviation of £25,000. Calculate, showing all working, a 90% confidence interval for the population average wage.

[7 marks]

c) Comment on the following statement: "For a fixed confidence level, the larger the sample size, the narrower the confidence interval is".

[4 marks]

d) Comment on the following statement: "Given a fixed confidence interval, the smaller the sample size, the lower the confidence level is".

[4 marks]

Total: 25 marks

Question 5:

An insurance company sold 1,000 one-year car insurance contracts, to policyholders spread across the country, at the price of £300 each. The observed average insurance claim that the insurance company had to pay during that year was £345 for the 1,000 contracts, with a standard deviation of £102.

The insurance company wants to test the hypothesis that the contracts are being underpriced.

Answer the following questions:

a) Write the appropriate test statistic to perform the desired hypothesis test and calculate the value of the test statistic.

[5 marks]

b) Perform the test at 1% confidence level.

[5 marks]

c) Based on your result above, explain if there is statistical evidence that the contracts are underpriced.

[5 marks]

d) Provide a 95% confidence interval for the true average claim value and compare your result with the price that the insurance company is charging for each contract.

[5 marks]

Total: 20 marks